WEEK 2: ASSIGNMENT 2

Q1 Which distribution is most appropriate for rare events occurring over a fixed interval?
a) Binomial
b) Normal
c) Poisson
d) Uniform

Q2 A fair coin is tossed twice. What is the probability of getting exactly one head?
a) 0.25
b) 0.50
c) 0.75
d) 1.00

Q3 Which distribution is commonly used to model the time between events in a Poisson process?
a) Binomial distribution
b) Exponential distribution
c) Uniform distribution
d) Chi-square distribution

Q4 Which of the following is a discrete probability distribution?
a) Normal distribution
b) Poisson distribution
c) Exponential distribution
d) Gamma distribution

Q5 If a binomial distribution has n = 10 and p = 0.4, what is the mean?
a) 2
b) 4
c) 6
d) 10

Q6 In a hypergeometric distribution, sampling is done:
a) Without replacement
b) With replacement
c) With constant probability
d) From infinite population

Q7 What will be the output of the following code:
"from scipy.stats import poisson
print(poisson.var(4))"

a) 2
b) 4
c) 8
d) 16

Q8 The mean and variance of a Poisson distribution are:
a) μ and μ^2
b) $\sqrt{\lambda}$ and λ
c) λ and $\sqrt{\lambda}$
d) λ and λ

Q9 Consider the following Python code:
"from scipy.stats import norm
print(norm.pdf(0))"

What will be the output?
a) 0
b) 0.25
c) 0.399
d) 1.000

Q10 Consider the following Python code:
"import scipy.stats as stats
print(round(stats.norm.cdf(0),2))"

What will be the output?
a) 0.00
b) 0.25
c) 0.50
d) 1.00

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